HSAN ELLOUZE

AI Engineering Student | AI Agents, LLM & RAG Systems
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Seeking a 6-month end-of-studies internship (PFE) starting February 2027 | Open to relocation (Berlin)

Summary

Final-year Software Engineering student (INSAT, expected graduation September 2027) building AI agents and LLM systems that run in production: a multi-agent workflow orchestrated with LangGraph, a configurable chatbot platform used by non-technical users, and bilingual RAG chatbots. Experienced in prompt engineering, API integrations and document AI, from ingestion and vector retrieval to evaluation across LLM providers. Grounded in the backend and fullstack engineering (FastAPI, Express.js, React/Next.js, Docker/CI-CD) needed to take an AI workflow from pilot to production.

Experience

AI Engineer Intern – Applicare, Canada Jun 2026 – Aug 2026
- Architected AskSahera, a bilingual (FR/EN) production **RAG** chatbot indexing ∼100 source documents
- Developed a dual-source ingestion pipeline (PubMed XML + GROBID PDF parsing) with IMRAD-aware chunking and a sliding-window fallback
- Built the RAG backend in Express.js with Qdrant vector storage and OpenAI embeddings, using Claude for generation; integrated Cloudflare R2 for storage

Mazdev – Tunis
Software Engineer Intern Jul 2025 – Sep 2025
- Engineered a production **chatbot platform** configured in YAML, letting non-technical users create 20+ bot variants without code
- Created a RAG engine (Pinecone) reaching 94% semantic accuracy across a large document set; optimized embeddings to cut latency 45%
- Established an automated testing framework for reliable releases, cutting the QA cycle 50%

Fullstack Developer Mar 2025 – Jun 2025
- Architected an AI-powered legal contract review platform automating document analysis for 40+ **enterprise clients**, cutting analysis time 60%
- Delivered a React/Next.js + FastAPI REST API system processing 500+ contract reviews per month at production scale
- Implemented a Docker CI/CD pipeline, cutting infra setup time 75%

Frontend Developer Intern – Steros, Sfax Jul 2024 – Aug 2024
- Built a responsive showcase website and interactive UI components, with a PHP/MySQL backend

Projects

KubeMind: AI-Driven Kubernetes Observability Platform (3-person capstone team) GitHub
- Designed a 5-layer **agentic** architecture (ingestion, AI agents, correlation/storage, presentation) automating incident diagnosis end to end
- Implemented 3 specialized **AI agents** with LangChain/LangGraph (logs, metrics, events) orchestrated as a LangGraph StateGraph workflow, feeding a correlation agent
- Developed a FastAPI backend and Next.js dashboard on a real-time ingestion pipeline (Prometheus, Loki, Kafka, MongoDB)
- Benchmarked output quality across 3 LLM providers (OpenAI, Gemini, Ollama) across 3 controlled incident scenarios

Multi-Tenant AI Chatbot Widget GitHub
- Designed a reusable, embeddable AI chatbot component for multi-tenant deployment across client websites via a single `<script>` tag
- Enforced Shadow DOM isolation and strict domain validation for secure multi-tenant use
- Integrated the backend with Hono, Supabase, and Upstash Redis; deployed on Railway and Cloudflare Pages

PromptCraft: AI NFT Generation & Auction Platform GitHub
- Created an end-to-end platform combining AI-generated artwork with a live auction marketplace
- Engineered a WebSocket/SSE bidding system supporting 100+ concurrent users, cutting latency ∼60%

Education

INSAT – National Institute of Applied Sciences and Technology Sep 2022 – Sep 2027 (expected)
Engineering Degree in Software Engineering (EQF Level 7 / Master's equivalent)

Skills

AI/ML: LLM Integration, LangChain, LangGraph, Prompt Engineering, RAG, Vector Databases (Qdrant, Pinecone), Embeddings
Languages: Python, JavaScript, TypeScript, Java, C#, PHP, SQL
Backend: FastAPI (Python), NestJS (JavaScript), Express.js (JavaScript), Hono (TypeScript), ASP.NET Core (C#)
Data & Observability: MongoDB, MySQL, Supabase, Redis (Upstash), Prometheus, Loki, Kafka
DevOps & Cloud: Docker, Kubernetes, GitHub Actions, CI/CD, Vercel, Railway, Cloudflare (Pages, R2)
Frontend: React, Next.js, Vue
Tools: Git, Linux

Activities

Event Mission Leader – Junior Enterprise INSAT 2024 – 2025
- Supervised a ∼40-member team over the 2024–2025 mandate, leading the organizing committee for 2 flagship events, including Forum INSAT Entreprise (1,500+ attendees, 40+ companies)

Operations Manager – WinterCup 7.0 2025
- Owned end-to-end logistics and event planning for WinterCup 7.0, an overnight coding contest with ∼300 participants, coordinating the on-site operations staff under time pressure